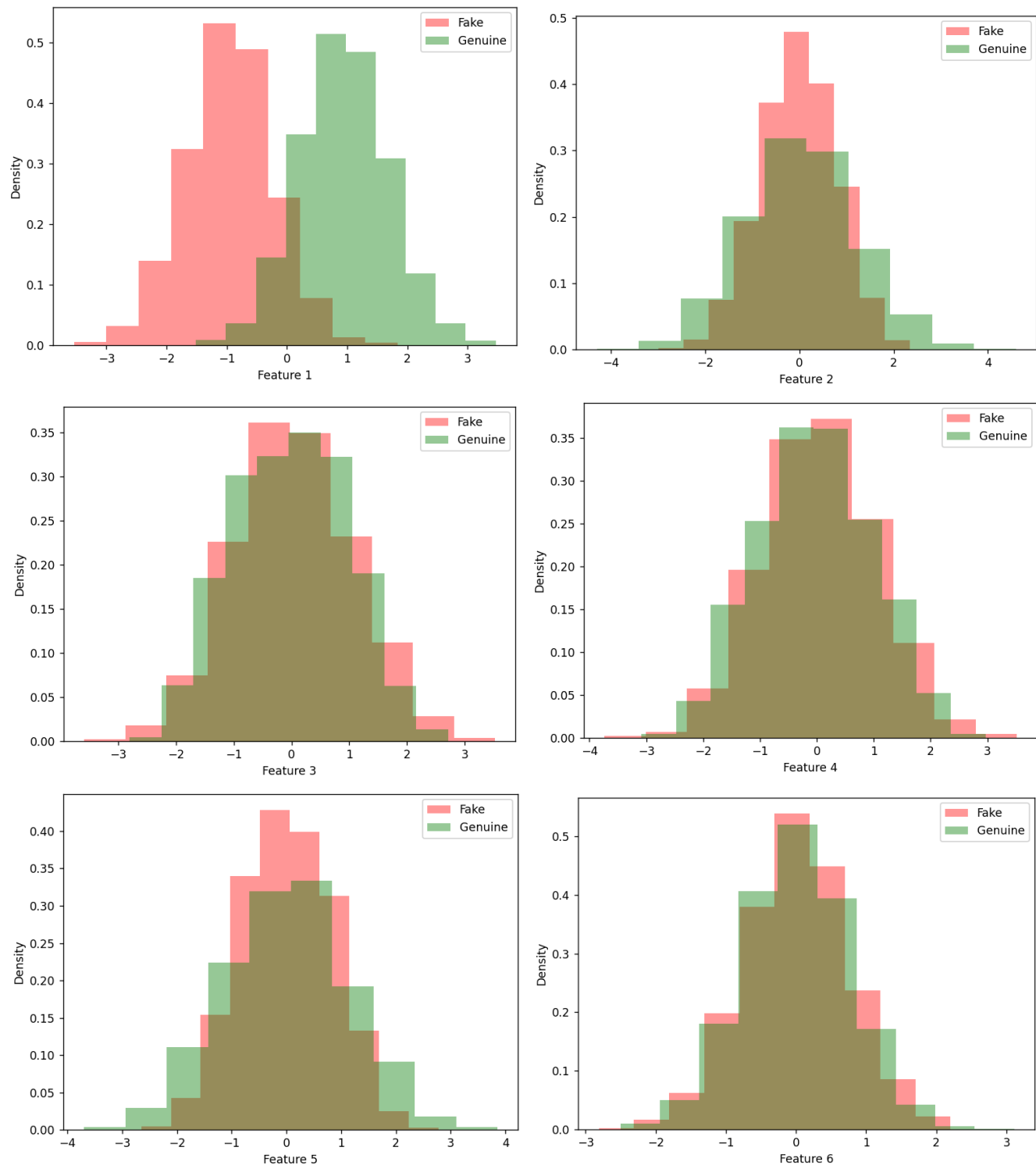


Lab. 3 - Project Report

Apply PCA and LDA to the project data. Start analyzing the effects of PCA on the features. Plot the histogram of the projected features for the 6 PCA directions, starting from the principal (largest variance). What do you observe? What are the effects on the class distribution? Can you spot the different clusters inside each class?



The first main components (Feature 1 and 2) seem to show a greater separation between the Fake (red) and Genuine (green) classes.

The following components show more overlapping distributions, suggesting that they contain less discriminating information.

PCA transforms the data into a new basis in which the first components have the greatest variance.

The classes maintain a normal distribution, but the overlap between them varies according to the direction considered.

In the first components, we note a possible internal structure in the classes, especially for the Fake class, which shows a more asymmetrical

distribution.

This may indicate the presence of subgroups or intra-class variations.

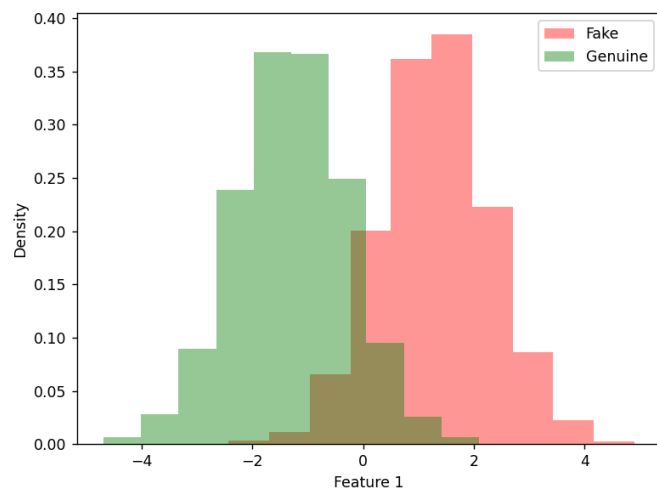
The PCA-transformed features show different distribution for individual features and different scatter plots. However, recall that PCA with 6 dimensions corresponds to just a rotation of the features. Have the clusters really changed? Have the classes really become “more separated”? (hint: the answer to both questions is no). This shows us that, while useful, exploratory analysis like histograms and scatter plots is often not sufficient to understand the global characteristics of our dataset, despite still giving some useful insights, as we will see in the next laboratories.

No, the clusters have not changed, they are still the same, but the distribution of the features has changed.

The classes have not become more separated, they are still the same.

For all the features the two classes overlapped in the same way, only the comparison with the first feature is seams to be more separated.

Apply LDA (1 dimensional, since we have just two classes), and compute the histogram of the projected LDA samples. What do you observe? Do the classes overlap over the found direction? Compared to the histograms of the 6 features you computed in Laboratory 2, is LDA finding a good direction with little class overlap? (Note: LDA is not increasing the separation between classes, it's finding a direction in the original feature space along which the classes are already well separated).



Compared to the histograms of the 6 features, LDA is finding a good direction with little class overlap.

The classes overlap less in the LDA histogram than in the histograms of the 6 features.

The first feature is the one that has the most separation between the two classes, and the LDA histogram is similar to the histogram of the first feature.

Try applying LDA for classification. Divide the dataset in model training and validation sets (you can reuse the previous function to split the dataset). Apply LDA, select the orientation that results in the projected mean of class True (label 1) being larger than the projected mean of class False (label 0), and select the threshold as in the previous sections, i.e., as the average of the projected class means. Compute the predictions on the validation data, and the corresponding error rate.

Results with Threshold = 0.018534362

Labels: [0 0 1 ... 0 0 0]

Predictions: [0 0 1 ... 0 0 0]

Number of errors: 186 (out of 2000 samples)

Error rate: 9.3%

Now try changing the value of the threshold. What do you observe? Can you find values that improve the classification accuracy?

Results with Threshold = 0.012356241

Labels: [0 0 1 ... 0 0 0]

Predictions: [0 0 1 ... 0 0 0]

Number of errors: 185 (out of 2000 samples)

Error rate: 9.2%

Results with Threshold = 0.009267181

Labels: [0 0 1 ... 0 0 0]

Predictions: [0 0 1 ... 0 0 0]

Number of errors: 185 (out of 2000 samples)
Error rate: 9.2%

Results with Threshold = 0.037068725

Labels: [0 0 1 ... 0 0 0]
Predictions: [0 0 1 ... 0 0 0]
Number of errors: 186 (out of 2000 samples)
Error rate: 9.3%

Results with Threshold = 0.004118747

Labels: [0 0 1 ... 0 0 0]
Predictions: [0 0 1 ... 0 0 0]
Number of errors: 184 (out of 2000 samples)
Error rate: 9.2%

Changing the value of the threshold does not improve the classification accuracy.
The error rate is always between 9.2% and 9.3%.
The best thresholds is 0.004118747 and , with an error rate of 9.2% and number of errors 184.
The worst threshold are 0.018534362 and 0.037068725, with an error rate of 9.3%.

Finally, try pre-processing the features with PCA. Apply PCA (estimated on the model training data only), and then classify the validation data with LDA. Analyze the performance as a function of the number of PCA dimensions m . What do you observe? Can you find values of m that improve the accuracy on the validation set? Is PCA beneficial for the task when combined with the LDA classifier?

Results with $m = 2$

Labels: [0 0 1 ... 0 0 0]
Predictions: [0 0 1 ... 0 0 0]
Number of errors: 185 (out of 2000 samples)
Error rate: 9.2%

Results with $m = 3$

Labels: [0 0 1 ... 0 0 0]
Predictions: [0 0 1 ... 0 0 0]
Number of errors: 185 (out of 2000 samples)
Error rate: 9.2%

Results with $m = 4$

Labels: [0 0 1 ... 0 0 0]
Predictions: [0 0 1 ... 0 0 0]
Number of errors: 185 (out of 2000 samples)
Error rate: 9.2%

Results with $m = 1$

Labels: [0 0 1 ... 0 0 0]
Predictions: [0 0 1 ... 0 0 0]
Number of errors: 187 (out of 2000 samples)
Error rate: 9.3%

Results with $m = 9$

Labels: [0 0 1 ... 0 0 0]
Predictions: [0 0 1 ... 0 0 0]
Number of errors: 186 (out of 2000 samples)
Error rate: 9.3%

Changing the number of PCA dimensions does not improve the accuracy on the validation set.
The error rate is always 9.2% or 9.3%.

The best number of PCA dimensions are 2, 3 and 4, with an error rate of 9.2% and the same number of errors.

The worst number of PCA dimensions is 1, with an error rate of 9.3%.